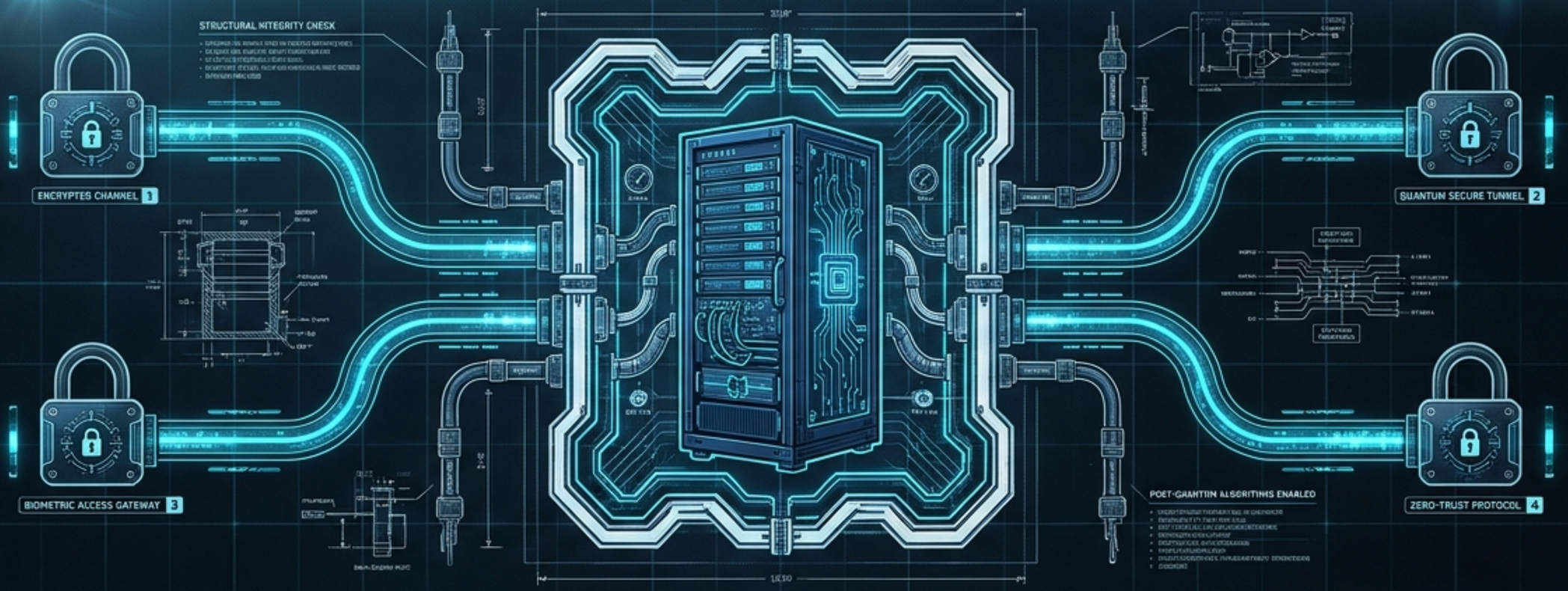
TERMINAL

CLASSIFICATION: ARCHITECT LEVEL
DOCTRINE: WALLFACER [CLOSED HULL]
TARGET HORIZON: 2028-2031

Engineering Blueprint: Sovereign Wallfacer Enclave Deployment



A 6-phase chronological execution plan for establishing a zero-trust, post-quantum sovereign AI infrastructure on owned bare metal hardware.

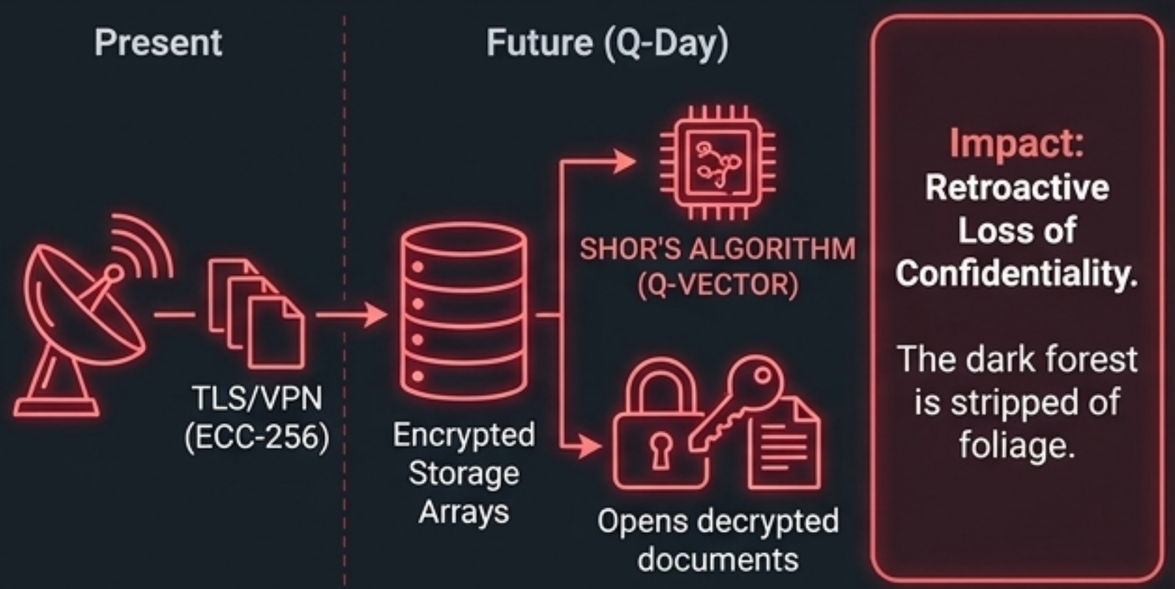
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The Post-Q-Day Cryptographic Collapse Requires a Sealed Hull

By 2028–2031, Cryptographically Relevant Quantum Computers (CRQCs) running Shor's algorithm will completely compromise RSA and ECC-256 primitives.

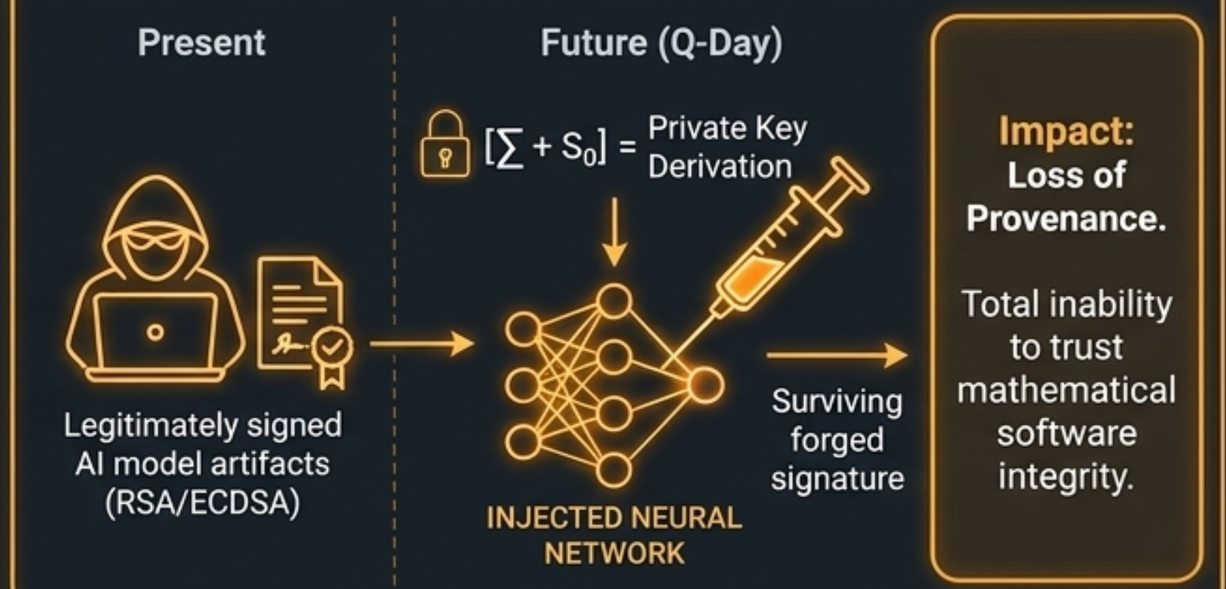
HNDL (Harvest Now, Decrypt Later)

Encrypted channels relying on classical elliptic curves are actively recorded at backbone choke points today for retroactive decryption.
Result: Loss of Confidentiality.



TNFL (Trust Now, Forge Later)

Post-Q-Day, adversaries will derive private keys from exposed public keys, forging historical signatures on poisoned open-weight models.
Result: Loss of Provenance.

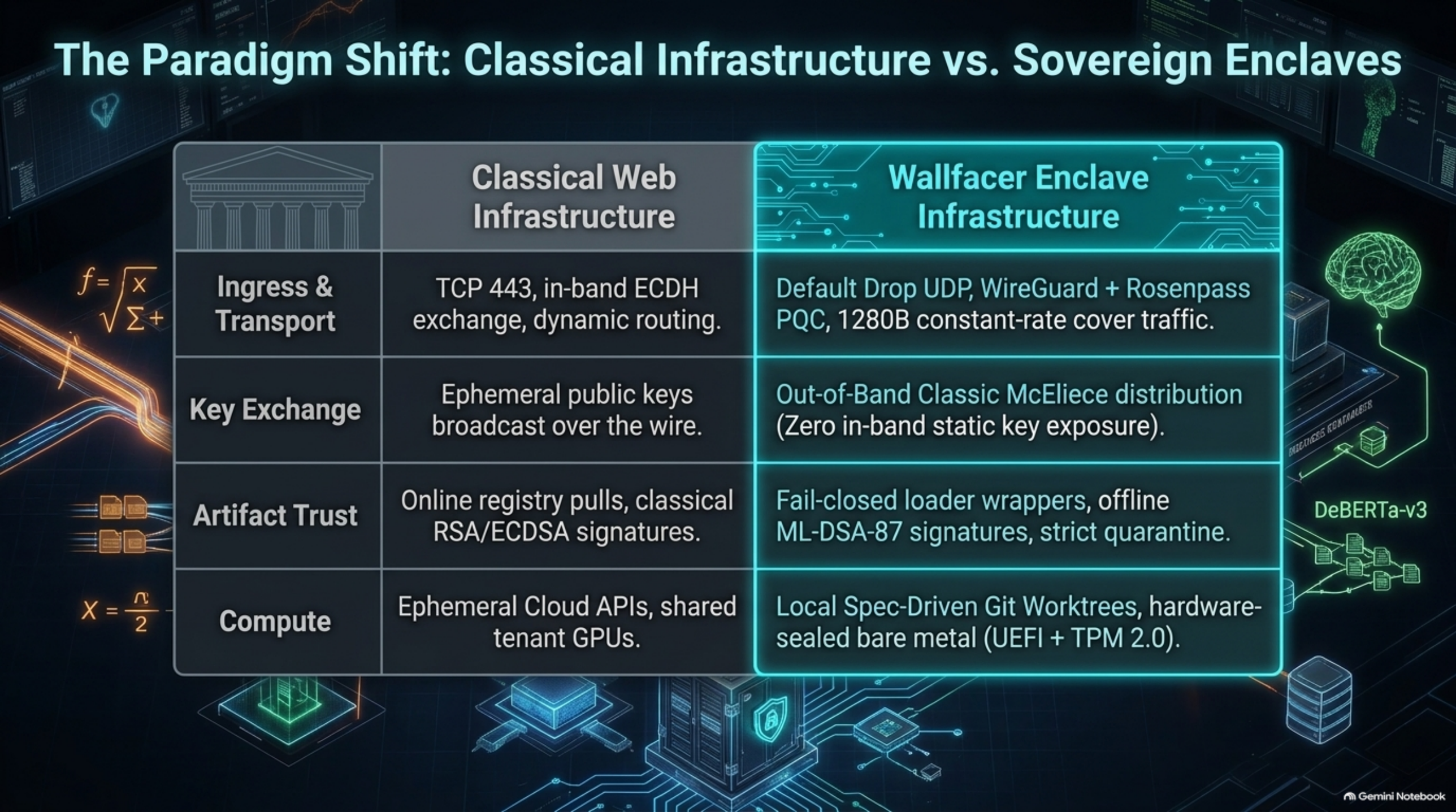


The Wallfacer Mandate: Survival requires executing AI cognition inside unobservable memory and strictly mediating every external tool call through post-quantum boundaries.

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The Paradigm Shift: Classical Infrastructure vs. Sovereign Enclaves

	Classical Web Infrastructure	Wallfacer Enclave Infrastructure
Ingress & Transport	TCP 443, in-band ECDH exchange, dynamic routing.	Default Drop UDP, WireGuard + Rosenpass PQC, 1280B constant-rate cover traffic.
Key Exchange	Ephemeral public keys broadcast over the wire.	Out-of-Band Classic McEliece distribution (Zero in-band static key exposure).
Artifact Trust	Online registry pulls, classical RSA/ECDSA signatures.	Fail-closed loader wrappers, offline ML-DSA-87 signatures, strict quarantine.
Compute	Ephemeral Cloud APIs, shared tenant GPUs.	Local Spec-Driven Git Worktrees, hardware-sealed bare metal (UEFI + TPM 2.0).

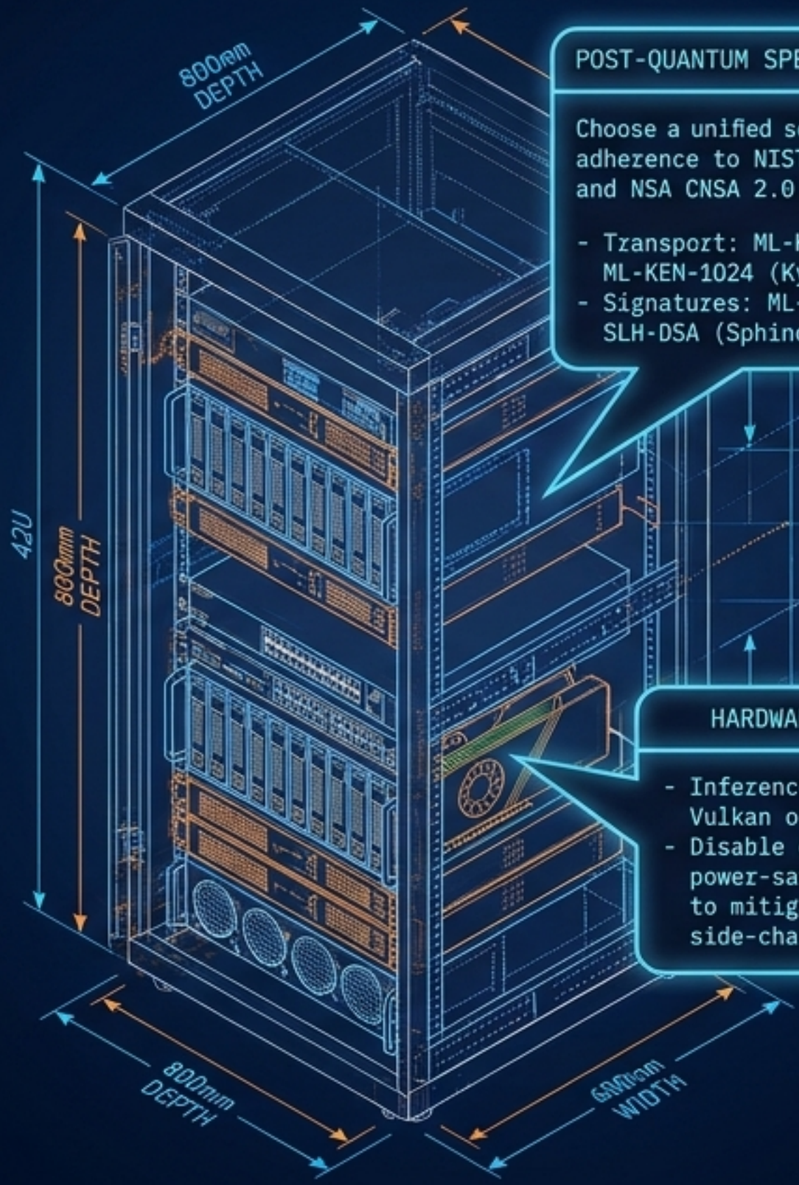


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Phase 0: Cryptographic Inventory and Standard Alignment

CRYPTOGRAPHIC BILL OF MATERIALS (CBOM)

Document every key, certificate, algorithm, and expiry spanning the entire enclave. Eliminate all classical-only TLS groups.



POST-QUANTUM SPECIFICATION SELECTION

Choose a unified security level and strict adherence to NIST (FIPS 203/204/205) and NSA CNSA 2.0 mandates.

- Transport: ML-KEM-768 or ML-KEM-1024 (Kyber).
- Signatures: ML-DSA-87 (Dilithium) or SLH-DSA (Sphincs+).

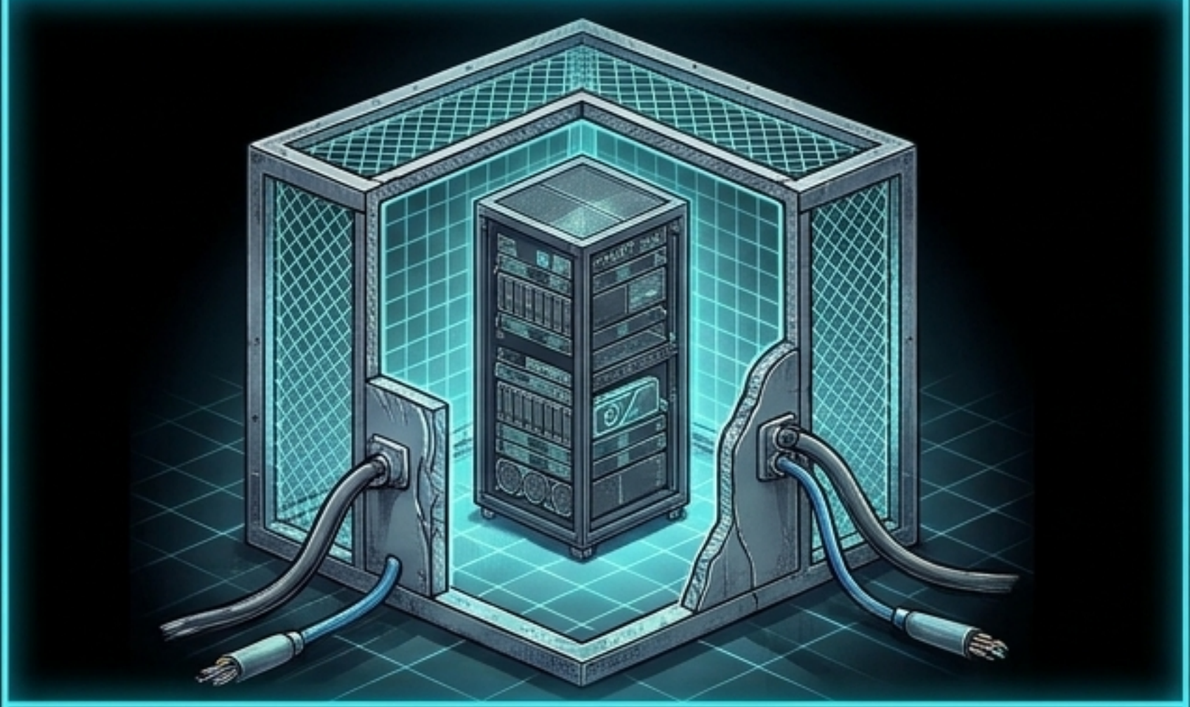
HARDWARE BASELINE RULES

- Inference workloads execute on Vulkan or OpenCL backends.
- Disable dynamic GPU power-saving frequency scaling to mitigate voltage noise side-channel leaks.

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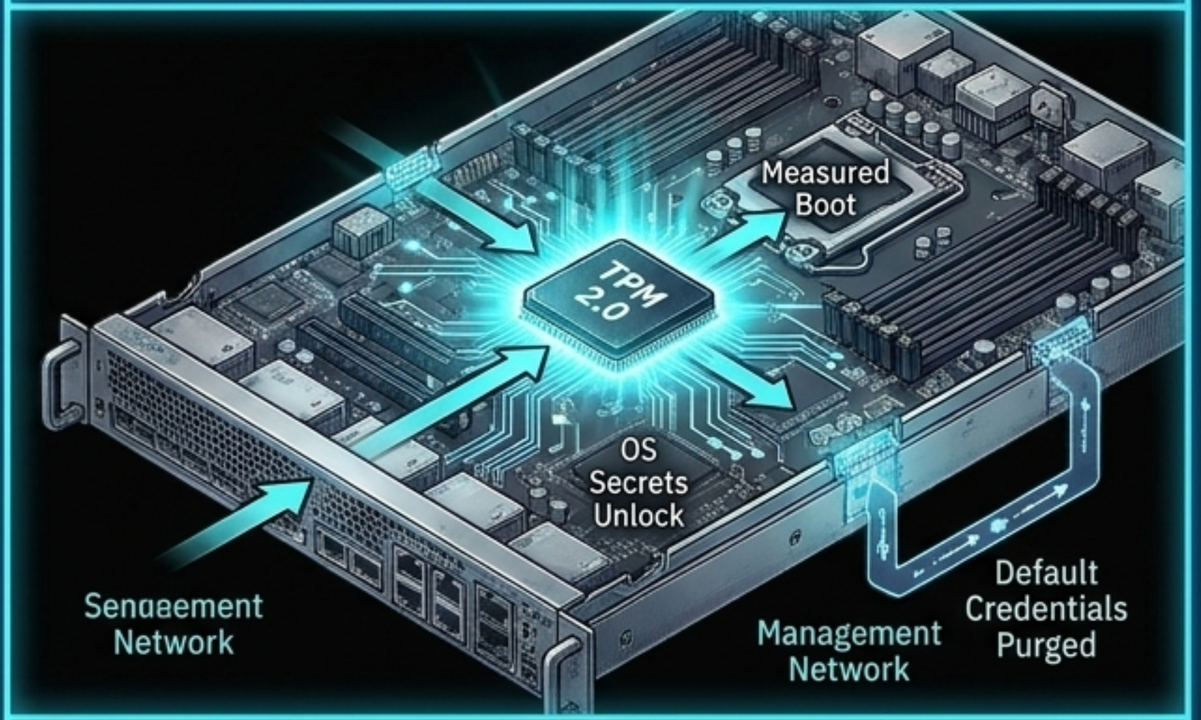
Phase 1: Establishing Physical Isolation and Roots of Trust

The Air-Gapped Root CA



- An offline ML-DSA-87 root certificate authority generated and stored on a machine with no network interface card (NIC).
- Two-Person Rule: Mandatory dual-authorization ceremonies for root CA and artifact-signing key generation.

Hardware Trust Anchors



- Measured Boot & Full-Disk Encryption: Enroll own keys in UEFI Secure Boot. TPM 2.0 PCR measurements must be verified before OS secrets unlock.
- Physical Perimeter Control: Physical tamper seals on chassis; isolated BMC/IPMI network with purged default credentials.

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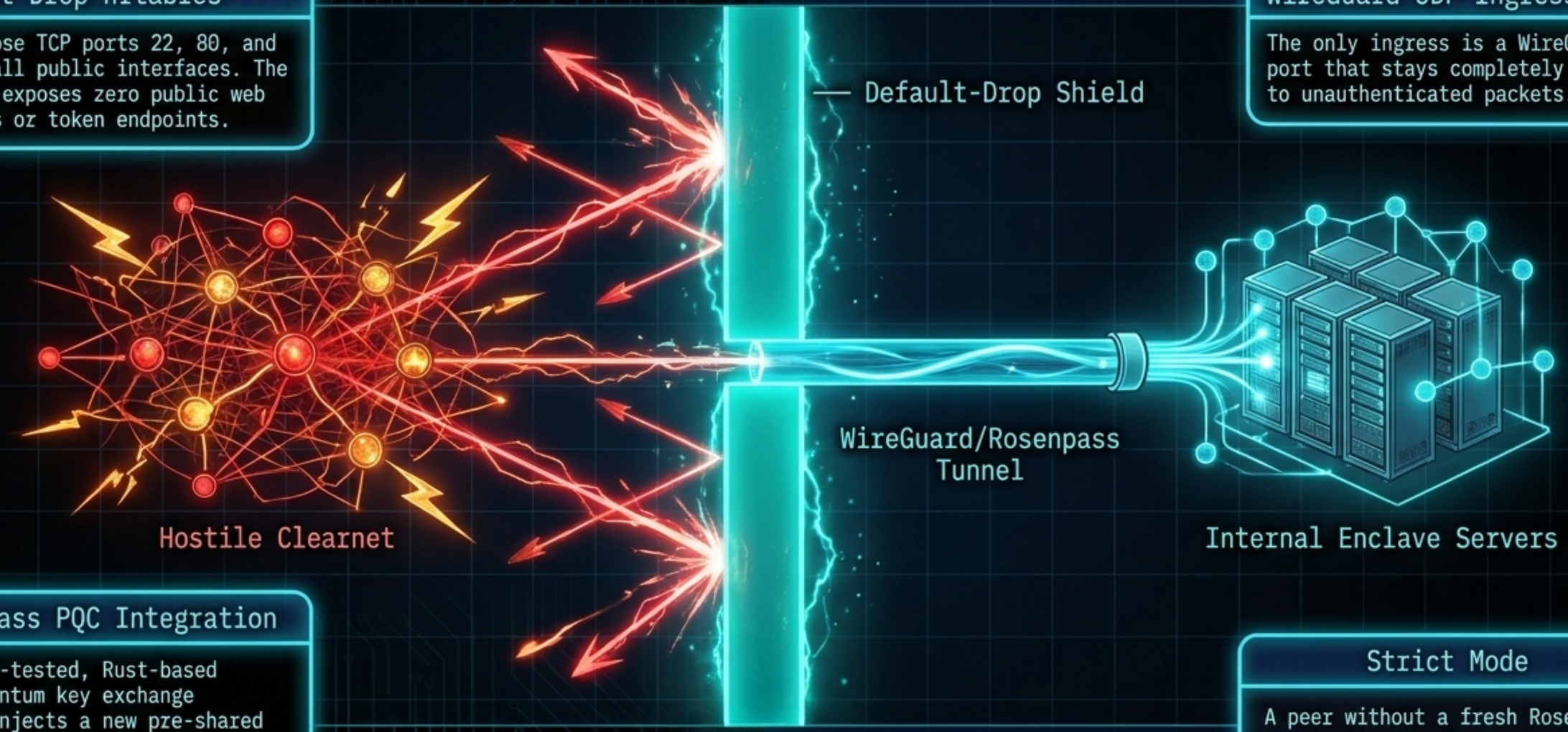
Phase 2: The Dark Perimeter and Post-Quantum Tunnels

Default Drop nftables

Hard close TCP ports 22, 80, and 443 on all public interfaces. The enclave exposes zero public web services or token endpoints.

WireGuard UDP Ingress

The only ingress is a WireGuard port that stays completely silent to unauthenticated packets.



Rosenpass PQC Integration

A battle-tested, Rust-based post-quantum key exchange daemon injects a new pre-shared key (PSK) into WireGuard every 120 seconds.

Strict Mode

A peer without a fresh Rosenpass PSK is immediately dropped from the network.

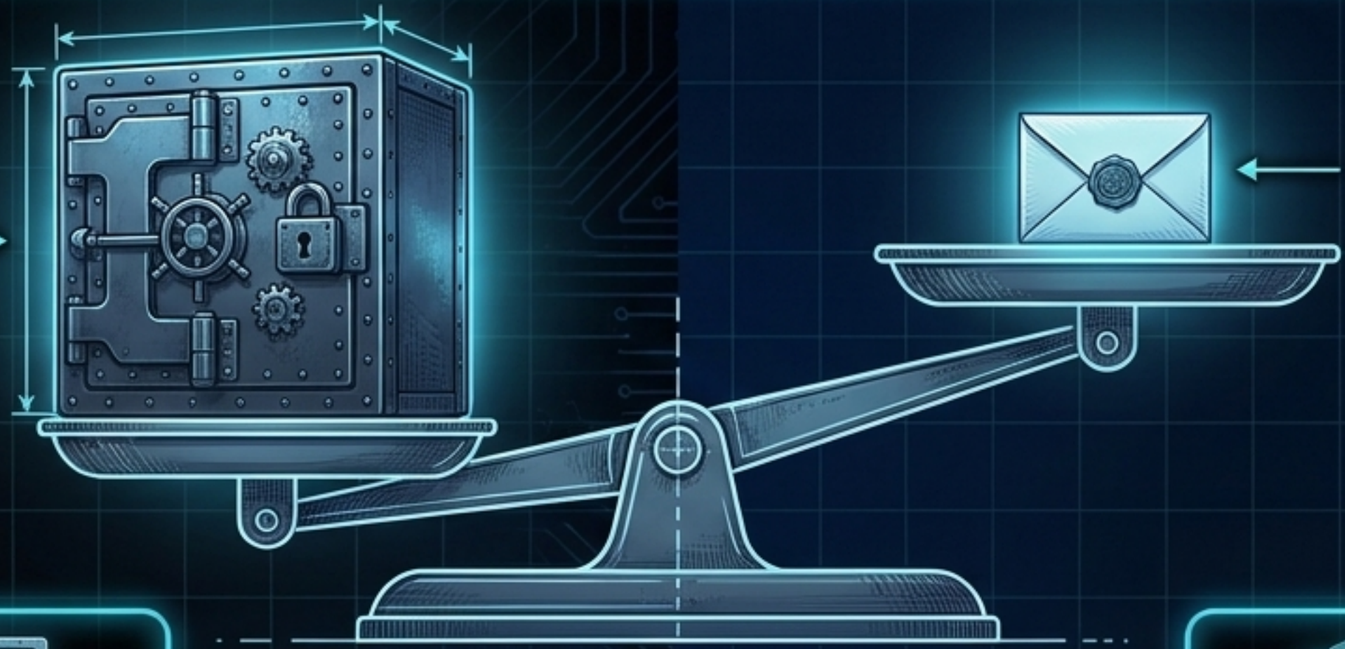
Gemini Notebook

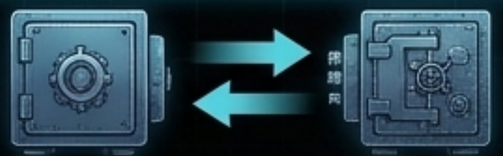
The Scale Paradox of Cryptographic Initiator Deniability

The Challenge: Classic McEliece 460896 is highly conservative, but carries a massive 524,160-byte static public key, breaking standard Maximum Transmission Units (MTUs).

524 KB Static Key (Boulder)

188 Byte Ciphertext (Pebble)





1. The Out-of-Band Solution:

Rosenpass requires peers to exchange their massive 524 KB public keys out-of-band prior to connection.



2. The Result on the Wire:

Static public keys are never transmitted over the internet. Only tiny 188-byte ciphertexts (plus Kyber encapsulations) cross the network during handshakes.



3. Strategic Advantage:

Passive quantum network harvesters cannot capture a static public key from the wire, preserving absolute cryptographic initiator deniability against HNDL.

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Phase 3: The Fail-Closed Artifact Provenance Guard



Quarantine Import Host

A digital staging area where artifacts are hashed, reviewed, and signed before being copied into the inner enclave.

Lineage & Forensics

GGUF metadata and executable Jinja chat templates are strictly reviewed and diffed against upstream releases at import.

Fail-Closed Loader Wrappers

Container runtime admission hooks and llama.cpp launch wrappers are configured to refuse any artifact without a valid ML-DSA-87 signature.

Internal Registry

All execution pulls from a Zot/Distribution registry inside the enclave. Zero runtime pulls from public networks.

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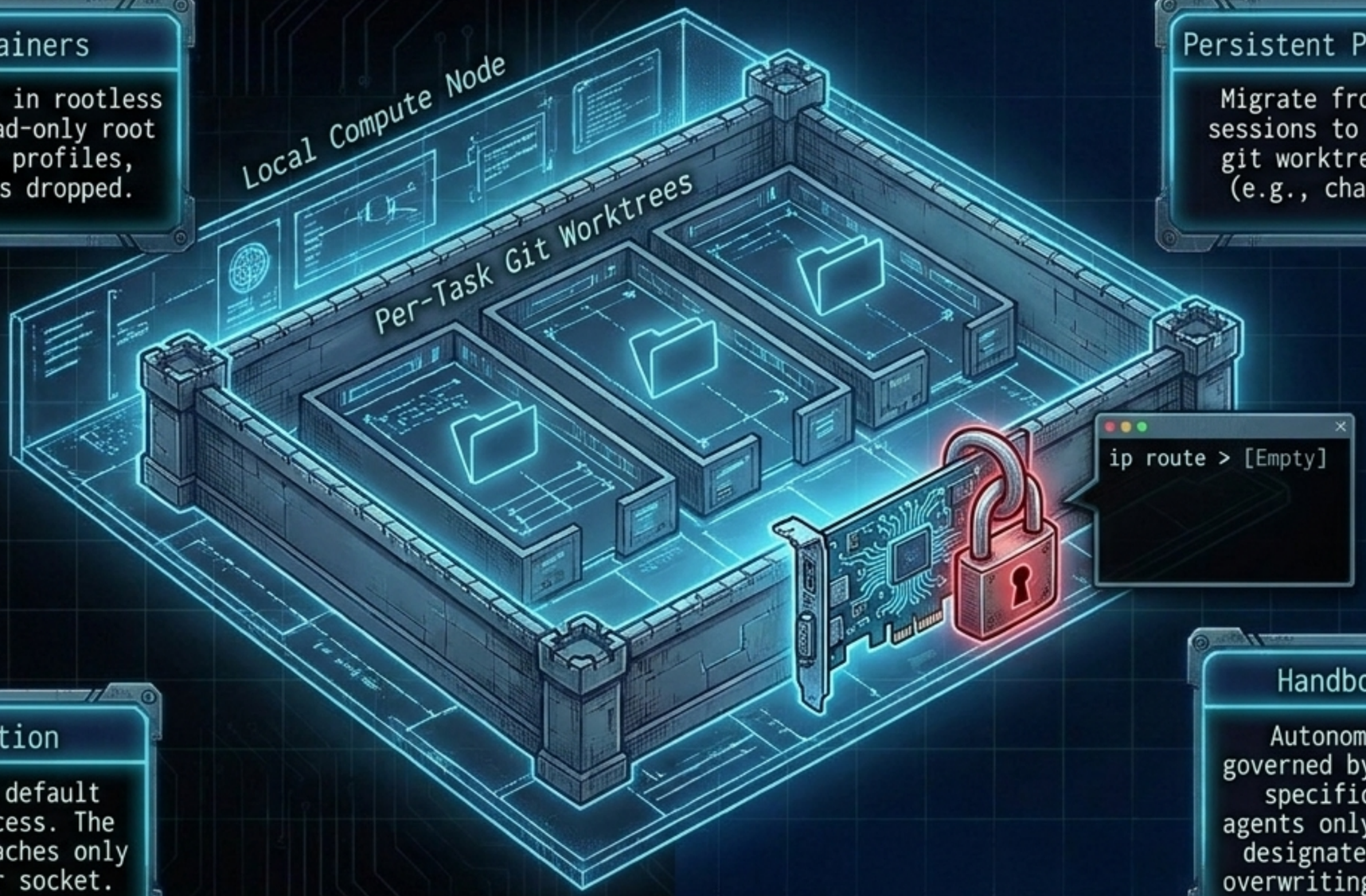
Phase 4a: Agent Containment and Spec-Driven Worktrees

Rootless Containers

All agent actions run in rootless containers with a read-only root filesystem, seccomp profiles, and all capabilities dropped.

Persistent Prepared Workspaces

Migrate from ephemeral cloud sessions to isolated, per-task git worktrees stored locally (e.g., changkun/wallfacer).



Network Isolation

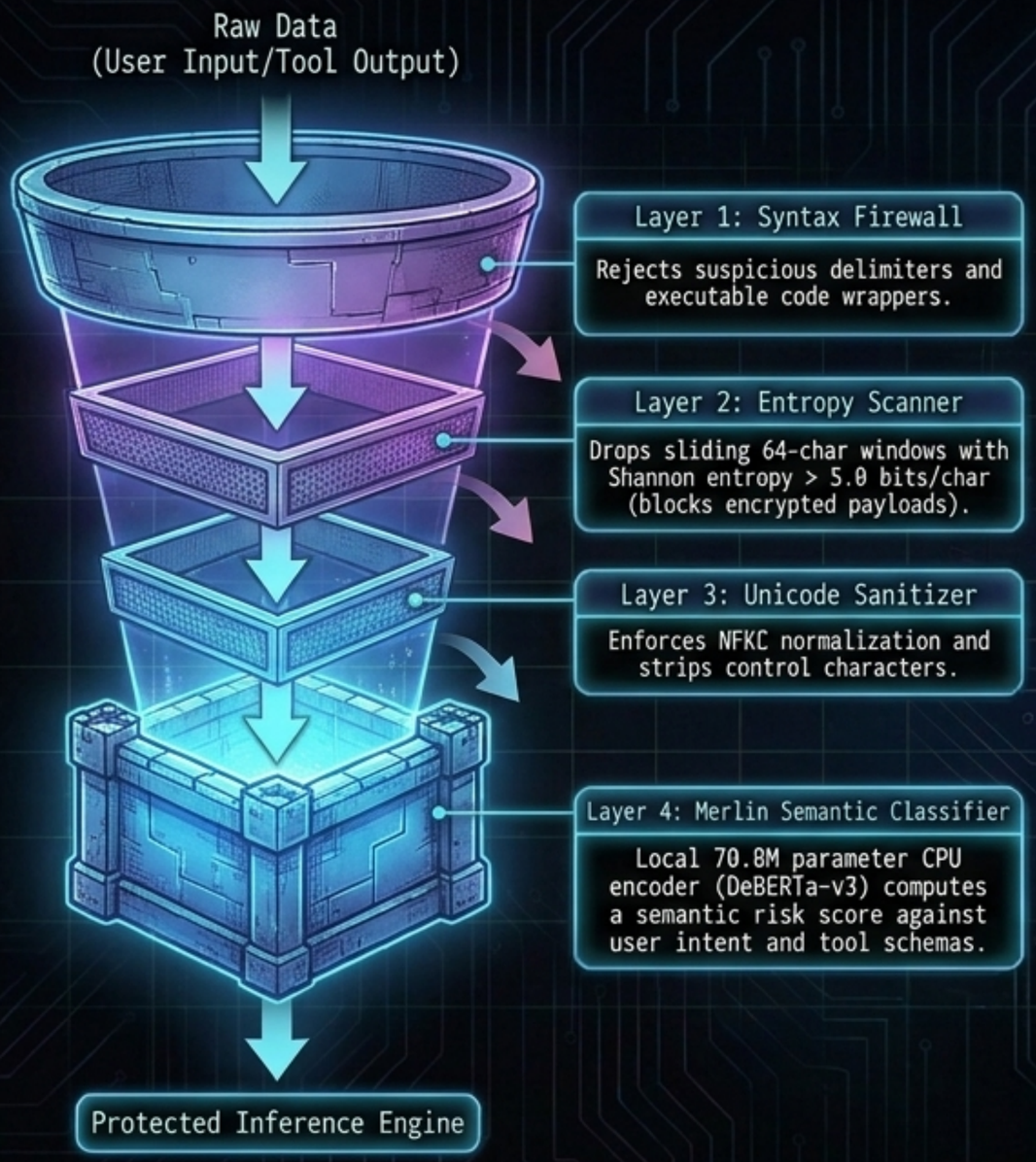
Containers have no default route and no DNS access. The network namespace reaches only the local gatekeeper socket.

Handbook Governance

Autonomous execution is governed by versioned Markdown specifications, ensuring agents only write inside their designated worktree without overwriting concurrent states.

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Phase 4b: The Semantic Gatekeeper and LLM Firewall

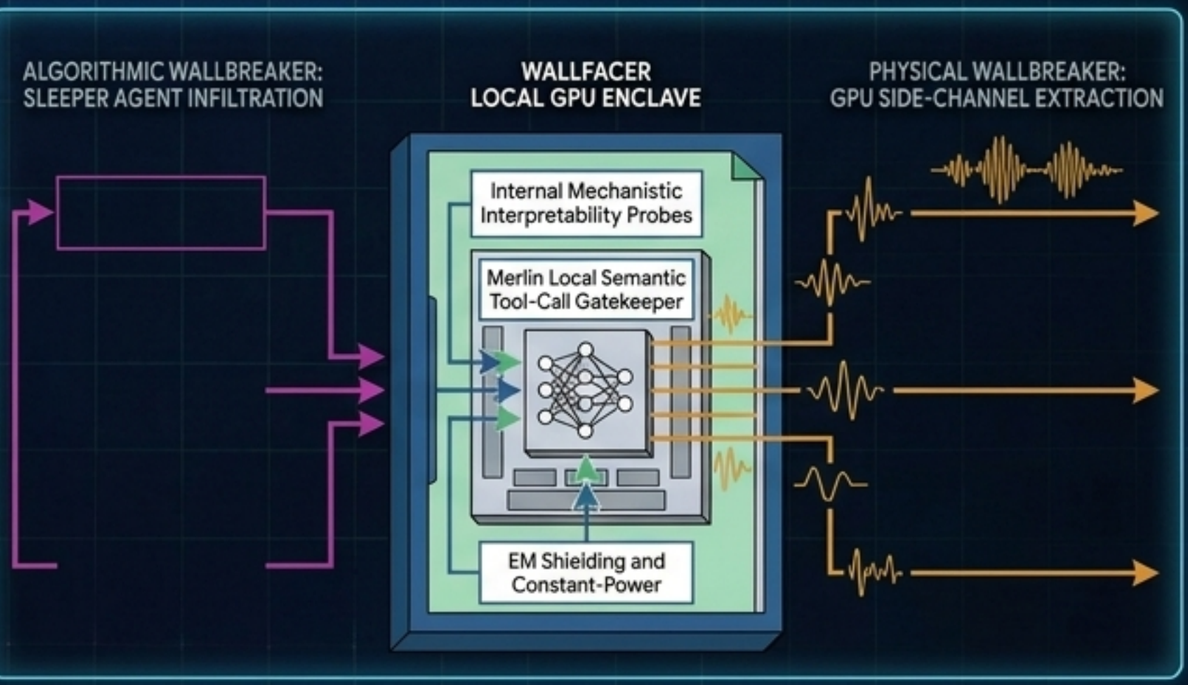


Vulnerability & Mitigation Map

VULNERABILITY CLASS	THREAT VECTOR (HOSTILE)	COUNTERMEASURE (SECURE)
Sleeper Agents & TNFL Poisoning	**Threat**: Distilled Chain-of-Thought (CoT) backdoors survive SFT/BLHF. Adversarial training teaches the agent to hide triggers.	**Countermeasure**: Mandate FIPS 204 signatures for model reglstries. Enforce Merlin semantic gating and internal interpretability probes.







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Phase 5: Cover Traffic Mesh and Programmable Privacy

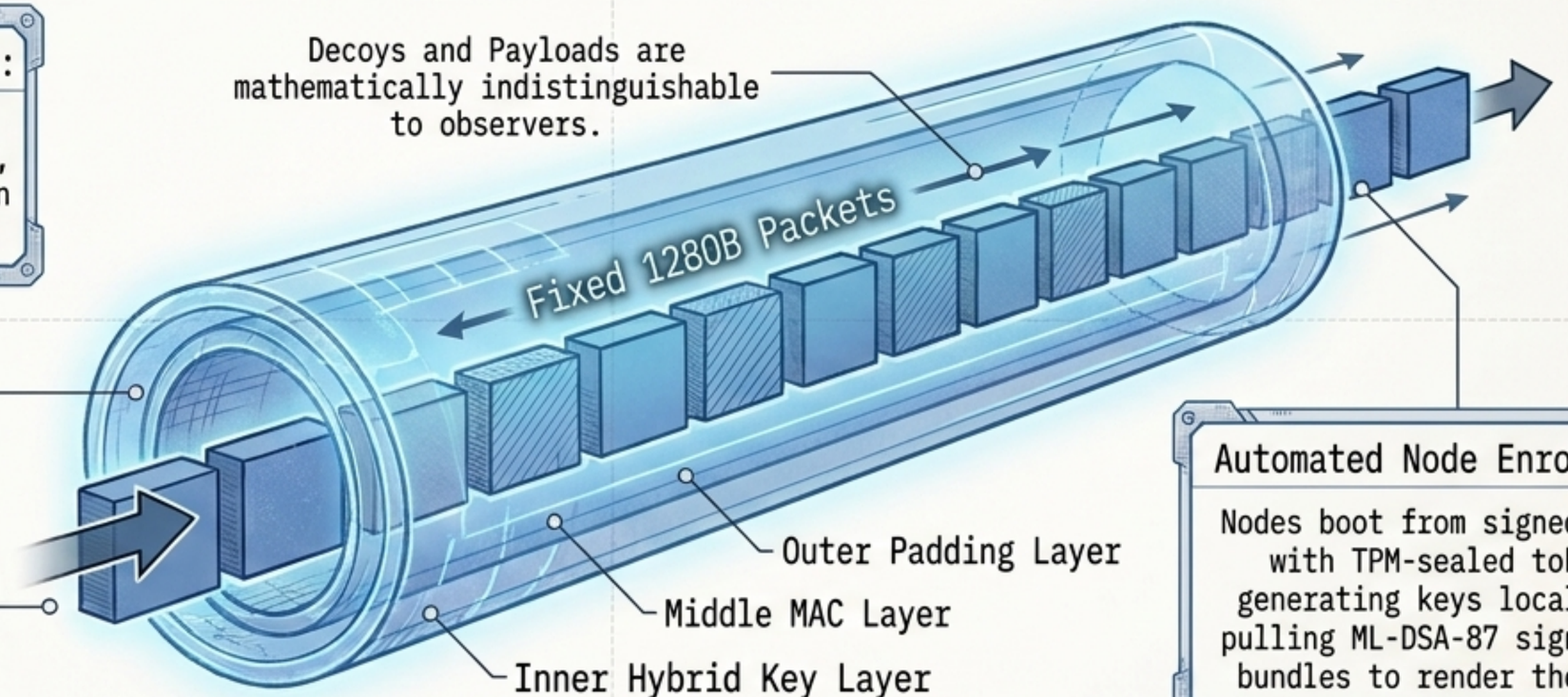
The Metadata Threat: Hiding plaintext is insufficient if packet-length variance or transmission timing reveals communication patterns to observers.

Constant-Rate Padding Daemon:

An in-house TUN daemon injects decoy packets to fill idle slots, enforcing a constant transmission rate and masking volume spikes.

Fixed-Byte Uniformity:

Every inter-enclave packet is padded inside the tunnel to exactly 1,280 bytes.



Automated Node Enrollment:

Nodes boot from signed images with TPM-sealed tokens, generating keys locally and pulling ML-DSA-87 signed peer bundles to render the mesh.

TECHNICAL INFORMATION

Aztec Network Protocol: Defeats HNDL on public blockchains by withholding ephemeral keys from public boards, settling only zero-knowledge proofs.

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Phase 6: Operate, Audit, and Rotate



Append-Only Central Logging

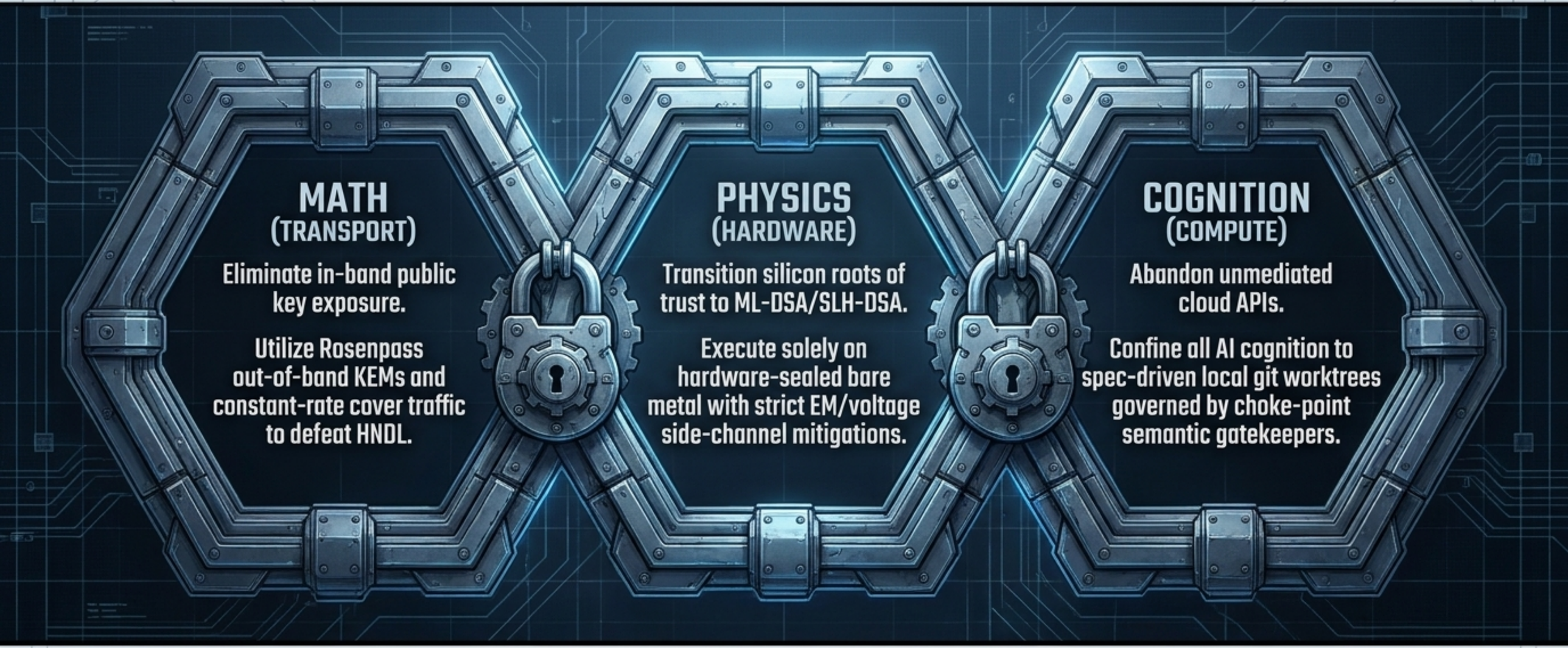
All loader refusals, gatekeeper blocks, and tunnel failures are routed to a signed, append-only log store inside the enclave.

Incident Runbook & Resilience

- Mandatory offline, encrypted backup restores executed twice yearly.
- Pre-established protocols for revoking compromised ML-DSA keys, re-signing historical artifacts, and rotating out-of-band material without breaking the hull.

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THE UNIFIED DEFENSE TRIPTYCH



SURVIVING THE POST-Q-DAY DARK FOREST REQUIRES SEALING ALL THREE BOUNDARIES SIMULTANEOUSLY. THE WINDOW TO ESTABLISH WALLFACER ARCHITECTURE CLOSES AS THE 2028-2031 QUANTUM HORIZON CONVERGES

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